

This report summarizes the performance of different neurons based on key metrics.

The metrics include KS Statistic, Pearson Correlation, and RMSE.

KS Statistic is...

A perfect Pearson correlation is 1. A correlation of 0 indicates no linear relationship.

RMSE is the square root of the average of squared differences between predicted and actual values. A 0 RMSE indicates no error.

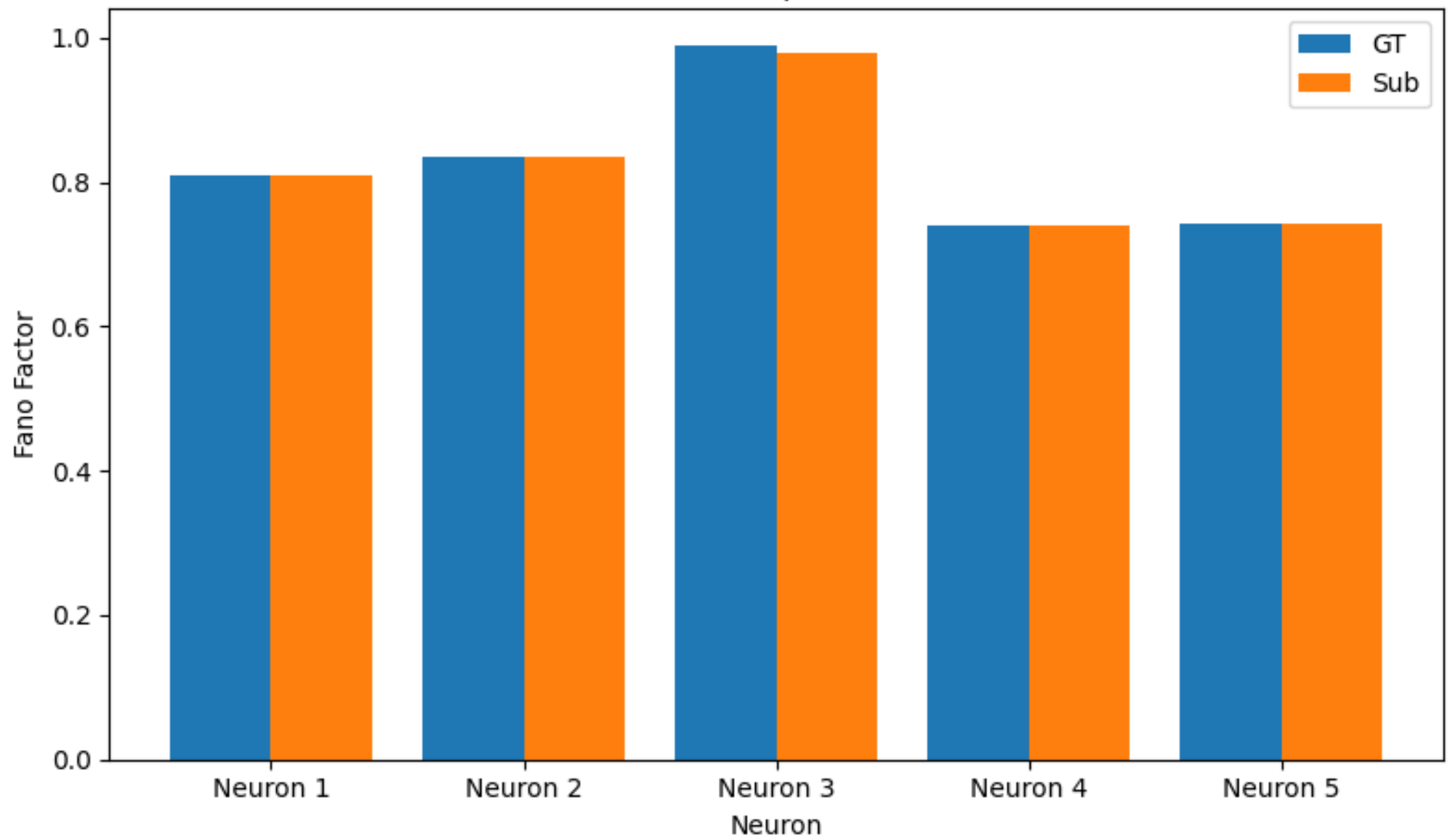
The Overall Score is a weighted combination of these metrics.

Summary Table of Spike Train Metrics

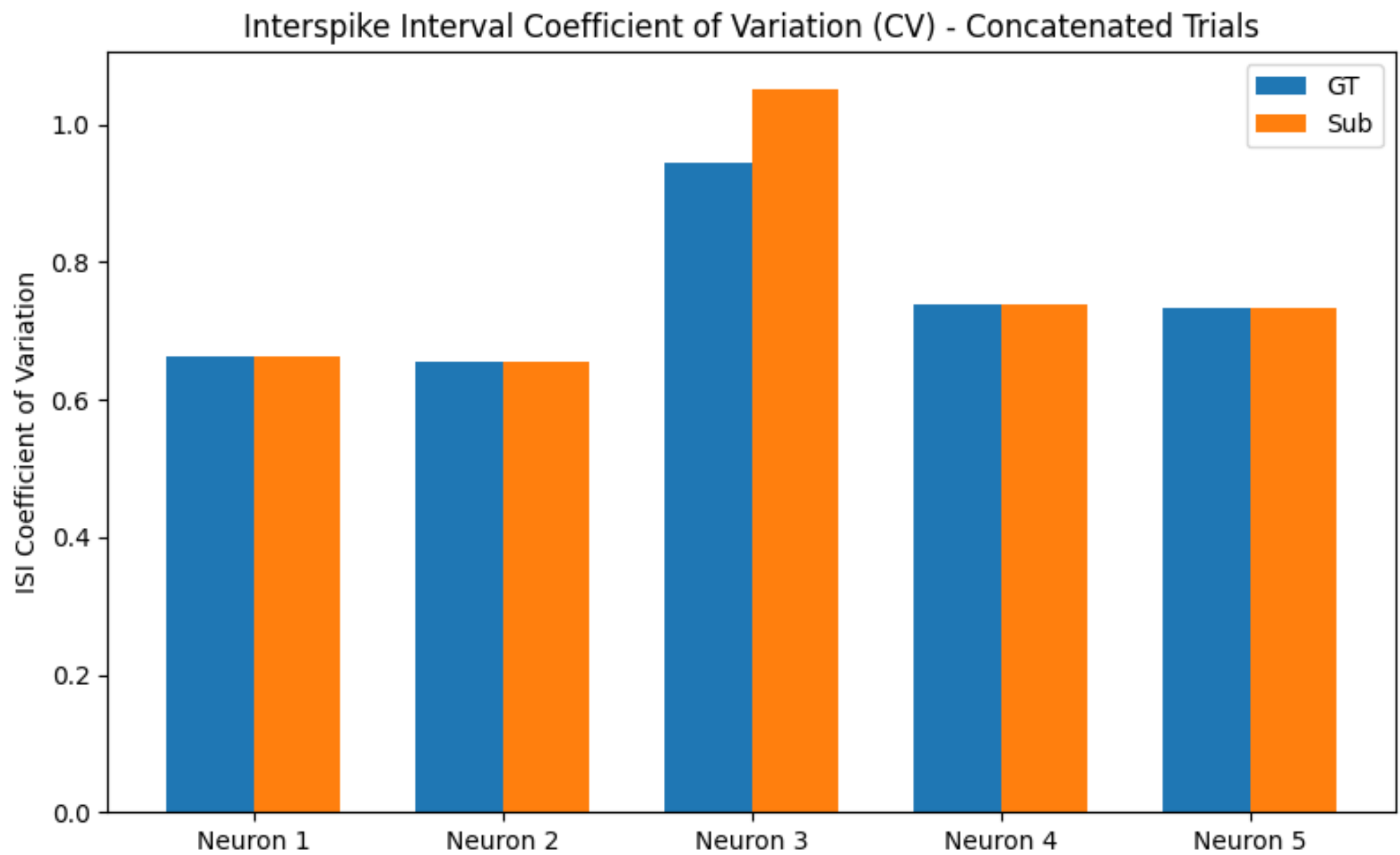
Neuron	KS Statistic	Pearson Correlation	RMSE	Overall Score
Neuron 1	0.0	1.0	0.0	100.0
Neuron 2	0.0	1.0	0.0	100.0
Neuron 3	2.056e-05	0.649	6.749e-05	0.0
Neuron 4	0.0	1.0	0.0	100.0
Neuron 5	5.556e-07	0.9997	1.054e-05	96.28

fano_factor.png

Fano Factor per Neuron

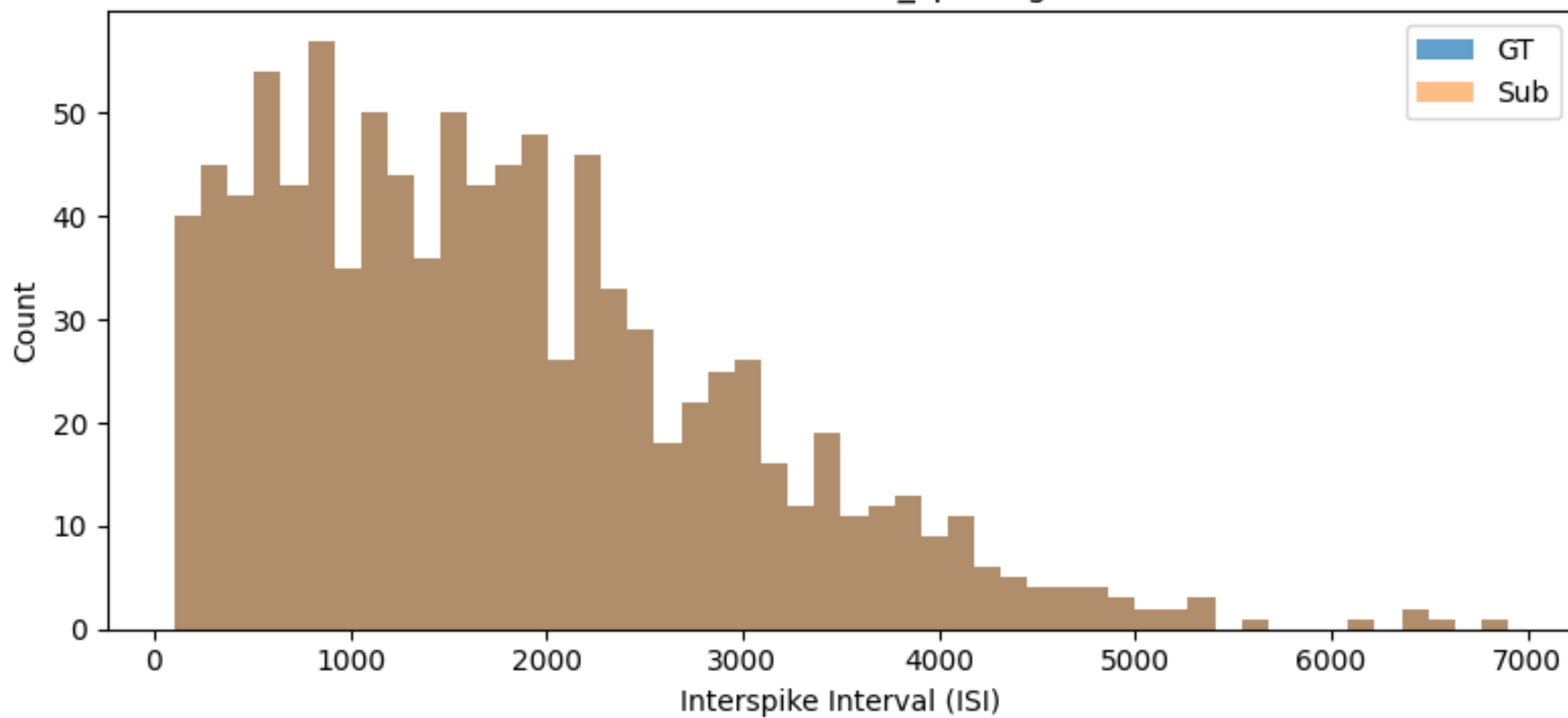


isi_cv.png



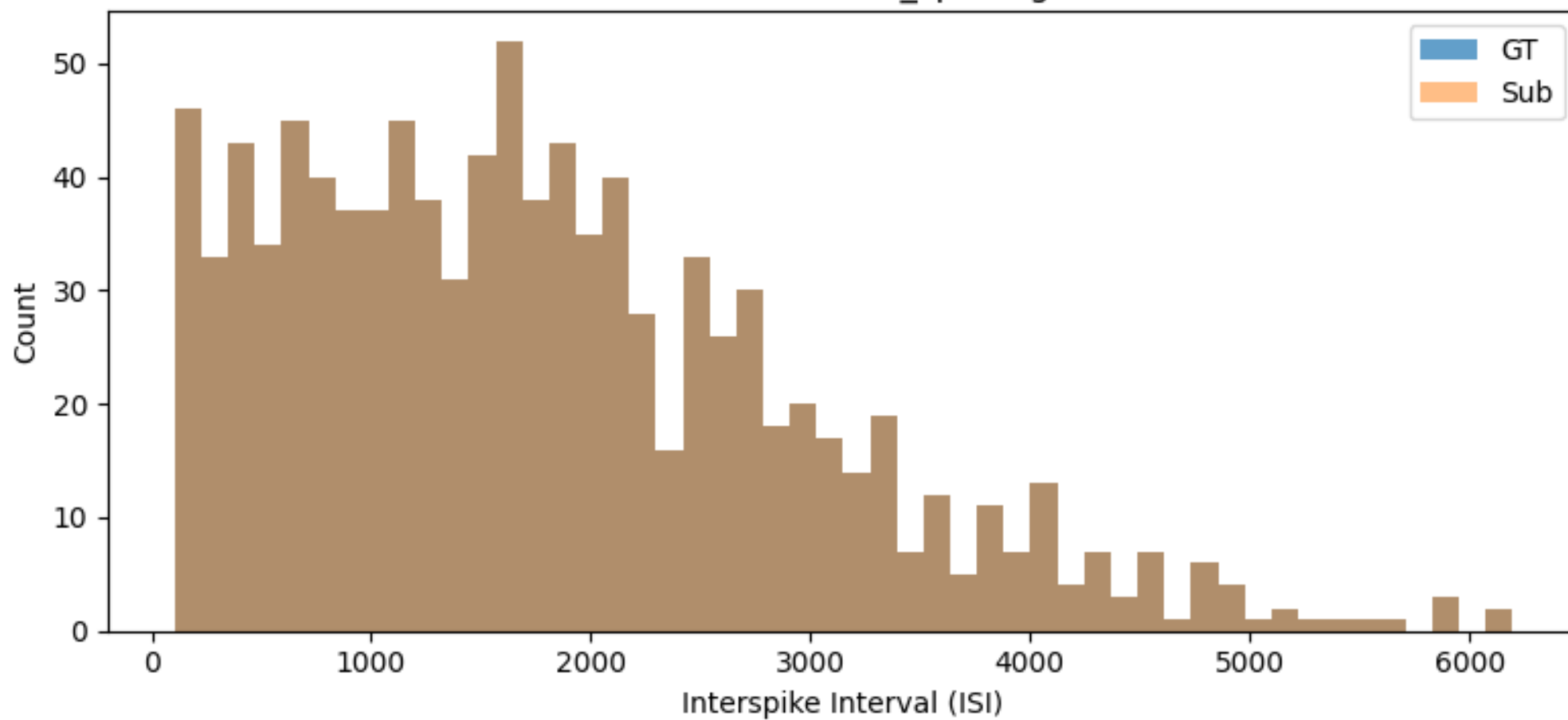
isi_dist_neuron1.png

ISI Distribution: neuron1_spike (gt vs sub)



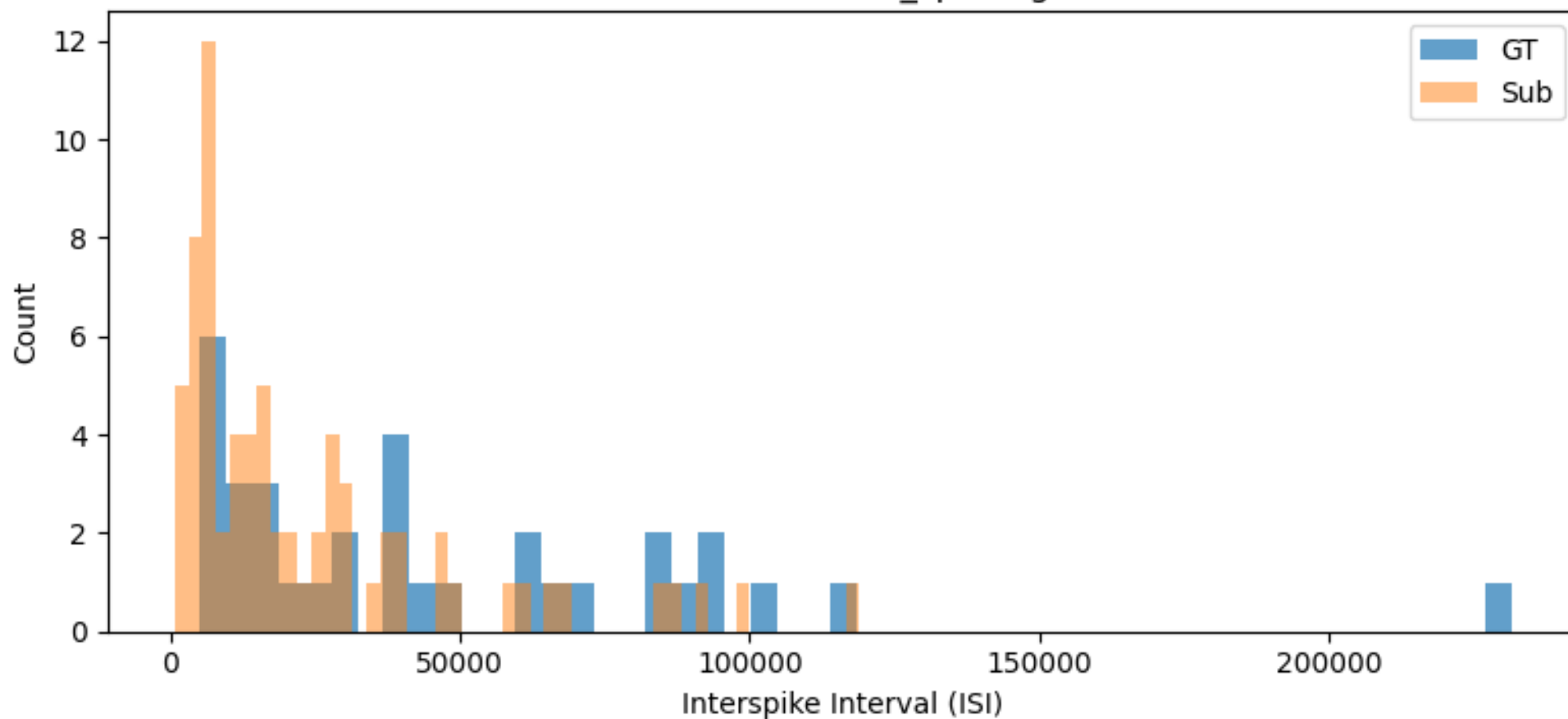
isi_dist_neuron2.png

ISI Distribution: neuron2_spike (gt vs sub)



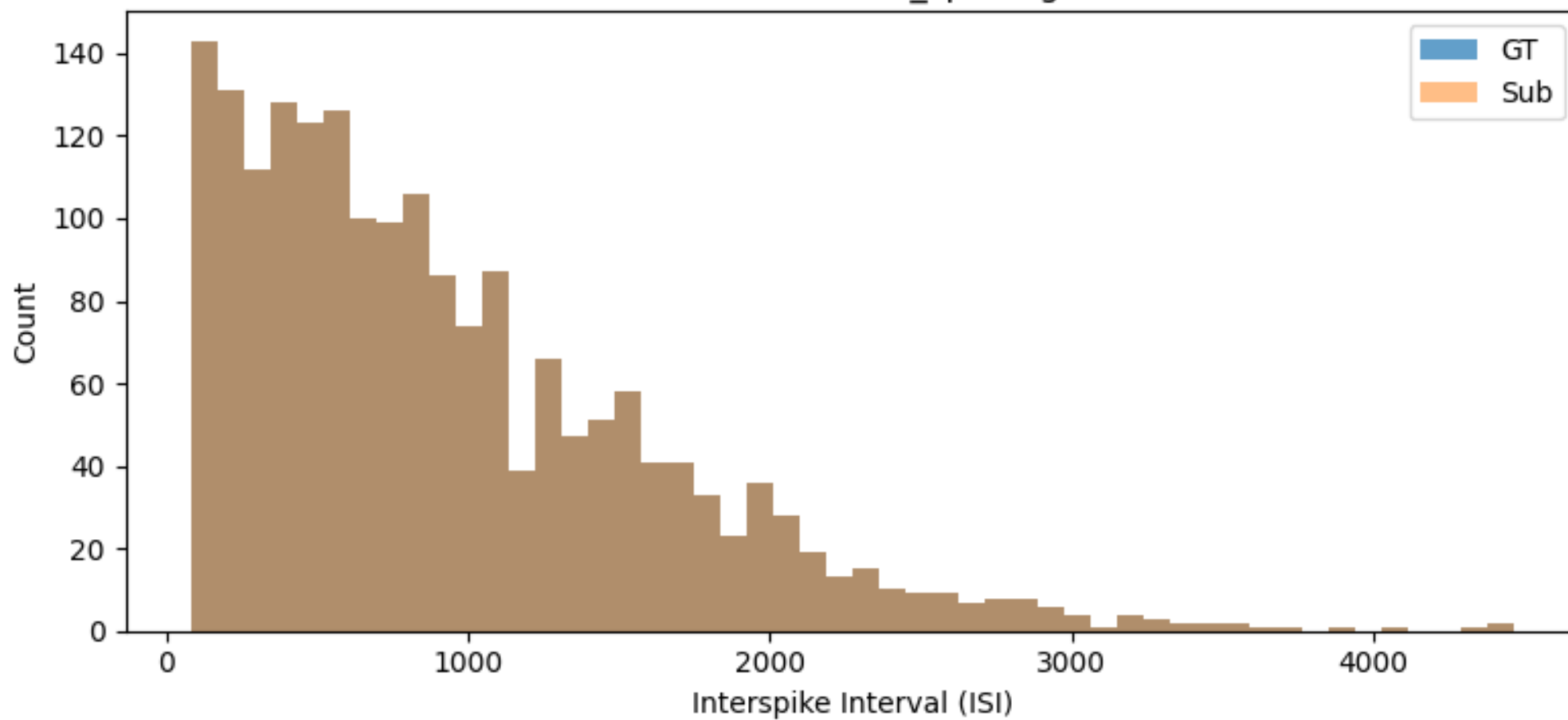
isi_dist_neuron3.png

ISI Distribution: neuron3_spike (gt vs sub)



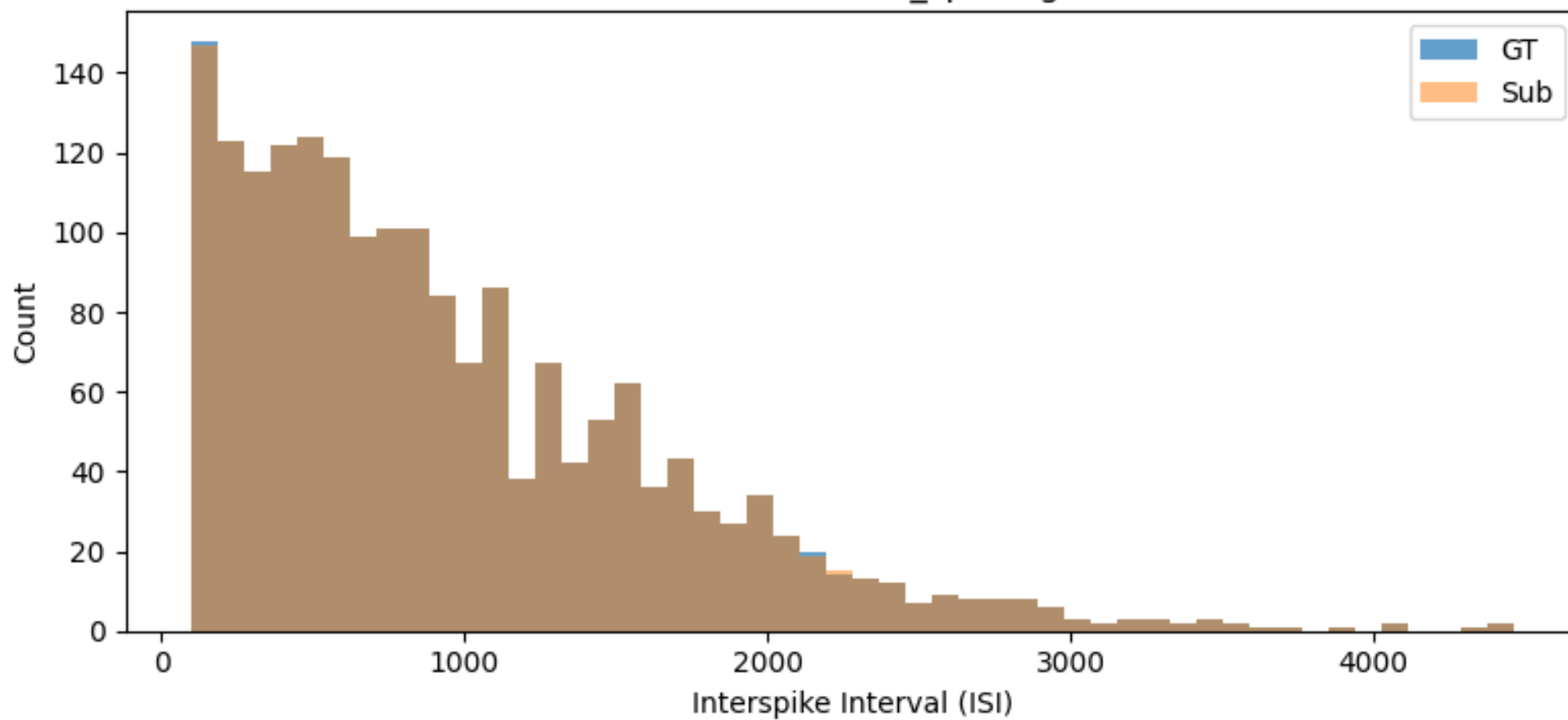
isi_dist_neuron4.png

ISI Distribution: neuron4_spike (gt vs sub)



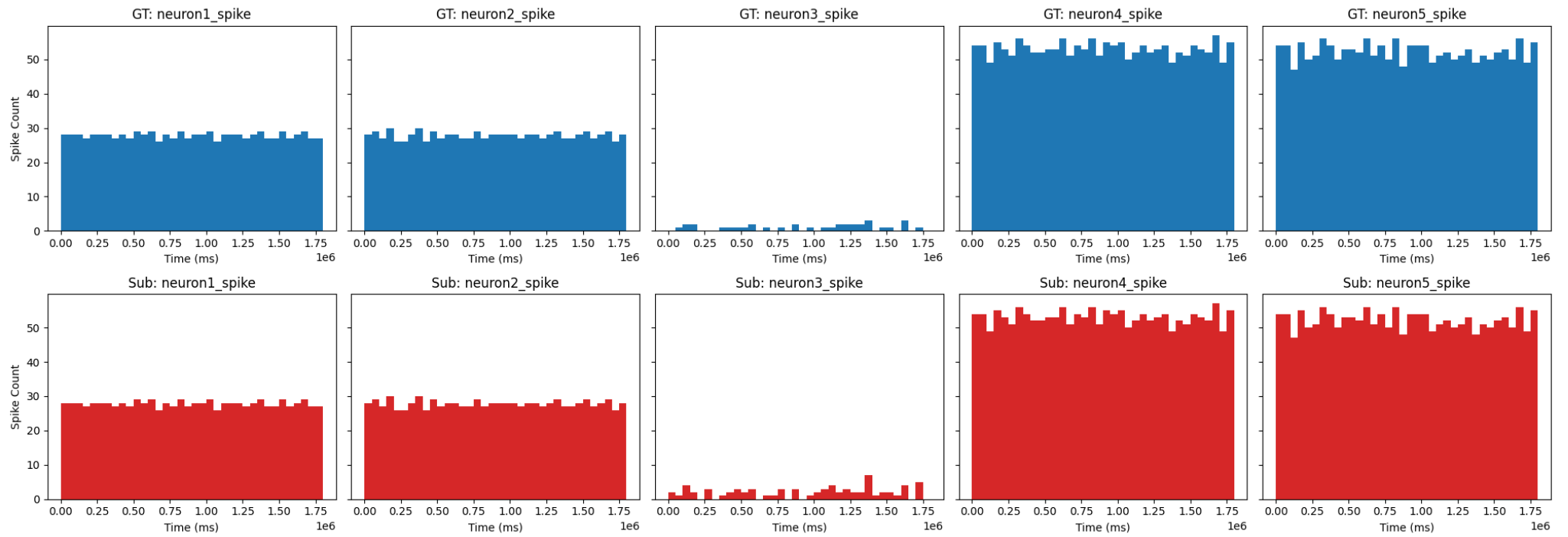
isi_dist_neuron5.png

ISI Distribution: neuron5_spike (gt vs sub)



peri_stim_histogram.png

Peri-Stimulus Time Histograms (PSTH)
Top: Ground Truth, Bottom: Submission



The figure is a raster plot with 'Time' on the x-axis (ranging from 0.00 to 1.75 million, with a $1e6$ multiplier) and 'Neuron' on the y-axis (listing neuron1_spike, neuron2_spike, neuron3_spike, neuron4_spike, and neuron5_spike). The plot shows vertical purple lines representing spikes. Neuron 3 has the highest density of spikes, with many clusters. Neuron 5 has only two spikes, one at approximately 1.22 million and another at 1.35 million. Neurons 1, 2, and 4 have no visible spikes in this time range.

The figure is a raster plot with 'Time' on the x-axis (ranging from 0.00 to 1.75 million seconds, with a multiplier of $1e6$) and 'Neuron' on the y-axis (listing neuron1_spike, neuron2_spike, neuron3_spike, neuron4_spike, and neuron5_spike). The plot displays vertical purple lines representing spikes. Neuron 3 has the highest density of spikes, occurring frequently throughout the time period. Neuron 4 has a moderate number of spikes, while Neuron 5 has the fewest, with only two spikes visible at approximately 1.23 million and 1.35 million seconds. Neurons 1 and 2 show no spikes in this visualization.